

# News Category Classification – Learning from Data

This repository contains the complete implementation of the News Category Classification final project for the Learning from Data course. The project covers data collection through web scraping, text preprocessing, feature engineering, model training, evaluation, and analysis.

## Project Overview

The goal of this project is to automatically classify news articles into predefined categories using machine learning techniques. Multiple traditional machine learning models and a neural network-based approach are implemented and compared within the Learning from Data framework. Key components of the project include: - Custom dataset collection via web scraping - Text preprocessing and feature extraction (BoW, TF-IDF, custom features) - Training and evaluation of multiple models - Bias-variance analysis and overfitting prevention.

## Repository Structure

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├── 01_eda.ipynb           # Exploratory Data Analysis
├── 02_preprocessing.ipynb # Text cleaning and feature engineering
├── 03_models.ipynb       # Model training and evaluation
├── clean_news_dataset.csv # Cleaned dataset
├── data2.csv             # Additional data file
├── tfidf_vectorizer.pkl   # Trained TF-IDF vectorizer
├── X_train_text.pkl
├── X_test_text.pkl
├── X_train_tfidf.pkl
├── X_test_tfidf.pkl
├── y_train.pkl
├── y_test.pkl
├── requirements.txt       # Python dependencies
└── README.md             # Project documentation
```

## Setup Instructions

1. Clone the Repository  
git clone <repository-url>  
cd <repository-folder>
2. Create a Virtual Environment  
Python -m venv venv  
Source venv/bin/activate
3. Install Dependencies  
All required Python packages are listed in requirements.txt
4. Run the Notebooks

Launch Jupyter Notebook and execute the notebooks in the following order:

Jupyter notebook

1. 01\_eda.ipynb
2. 02\_preprocessing.ipynb
3. 03\_models.ipynb

## **Data Collection**

The dataset was created by scraping publicly available news websites in compliance with their terms of service. Raw data and preprocessing steps are fully reproducible using the provided notebooks.

Reproducibility

- All experiments are fully reproducible using the provided notebooks.
- Random seeds are fixed where applicable.
- Trained artifacts (e.g., TF-IDF vectorizer) are included for consistency.

## **Author**

This project was developed as part of the Learning from Data course.

## **License**

This project is for academic use only